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Design and optimization of phase frequency detector through Taguchi and ANOVA statistical techniques for fast settling low power frequency synthesizer

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ABSTRACT

In this work, a novel phase frequency detector (PFD) architecture using pass transistor logic is proposed. The circuit does not have a reset path, resulting in the elimination of blind zone and dead zone. The ϕ-V characteristics of the PFD were found to have better linearity across the range of $-\pi$ to π due to the absence of blind and dead zones. The Taguchi and ANOVA statistical techniques were used to optimize the PFD. The optimized PFD exhibited a phase noise of −142.24 dBc/Hz, consumed 5.64 μW of power and had a maximum operating frequency of 5.25 GHz, and a delay of 10.65 ps. Using this PFD, a GHz-range synthesizer was designed, and its performance characteristics were obtained from circuit simulations using CADENCE Virtuoso. The synthesizer had a power consumption of 4.25 mW at a supply of 1.8 V, achieved a lock time of 2.95 μs, and could generate frequencies ranging from 0.1 GHz to 4.75 GHz while occupying a chip area of 0.013 mm^2. Moreover, the work introduced a new figure of merit, FoM. The synthesizer has potential applications in various devices such as radio receivers, televisions, mobile phones, satellite receivers, and GPS systems.

1. Introduction

Advancements in communication devices and circuit design techniques is leading to a heightened demand for cost-effective construction of high-performance frequency synthesizers. Frequency synthesizer is a crucial component for generating high-precision frequency signals from a reference clock, and is utilized in various microprocessors, micro-controllers, and wireless communication systems [1–3]. The critical design aspects of a frequency synthesizer are characterized by low phase noise, broad frequency coverage, smaller area, and lower supply voltage. The construction of frequency synthesizers requires components such as a phase frequency detector (PFD), charge pump (CP), loop filter (LF), differential ring oscillator (DRO), and frequency divider (FD) [4–6]. Faster locking PLLs/Frequency synthesizers are necessary to support the high-frequency operations of communication and electronic devices [7–10]. Fig. 1 shows the fundamental block diagram of frequency synthesizer.

The classification of a frequency synthesizer as analog or digital is dependent on the type of PFD, LF, and DRO utilized. Additionally, synthesizers are also categorized as Integer-N or Fractional-N. The advantage of Integer-N synthesizers is their lower cost and simpler design [11–13]. Applications that require small frequency step sizes and low phase noise may benefit from Fractional-N synthesizers, although they are more complex [14–16]. Researchers have been striving to design and construct efficient frequency synthesizers in recent years [17, 18]. The PFD is an important component of the frequency synthesizer as it compares the reference signal to the feedback signal. The PLL attains phase synchronization and is considered “locked” when the inputs of PFD are in phase alignment.

The charge pump device takes outputs from the PFD and utilizes these signals to adjust the control voltage, either increasing or decreasing it. A low-pass filter is employed to eliminate high-frequency components from the PFD outputs. The VCO adjusts its output frequency in accordance with the signals from the charge pump [19].

To create frequencies of interest, wireless applications like Satellite, Infrared, Radio, Microwave, Wi-Fi, etc. need some kind of frequency synthesizer. This paragraph discusses various frequency synthesizers reported in different research works. The authors in [1] describe the PLL for minimizing the reference spurs and creates frequencies between 5.7 to 6 GHz. The research work in [2] discusses mixed PLL for reduced noise and quick settling time. The frequency range produced by this

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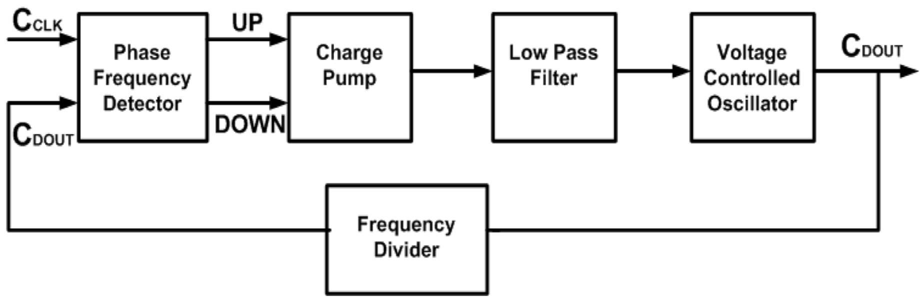


Fig. 1. Frequency synthesizer block diagram.

circuit is 0.025 to 1.6 GHz. The synthesizer shown in [3] is capable of producing frequencies between 10 MHz and 1.5 GHz. In [20], a type-I synthesizer architecture with no inductor was suggested for 2.4 GHz RF applications. The authors in [21–24] suggest synthesizer designs that aim to achieve low jitter, a broader range of frequencies frequency, and moderate power usage. In [25], a synthesizer using a voltage-controlled oscillator with inductor and capacitor and a static D-flipflop is described, which has reduced phase noise and power usage. The auto-calibration method for quick frequency calibration of VCO is provided to obtain a frequency range of 1.9–3.8 GHz [26]. A fast-settling PLL circuit generating frequencies from 130 MHz to 1 GHz is presented in [27].

The proposed research presents a novel architecture for a phase frequency detector that employs pass transistor logic, ensuring lower power consumption. In this design, the reset path is eliminated, resulting in a total of only 8 transistors being used. Furthermore, the proposed architecture is free of any dead zones and operates at high frequency. A frequency synthesizer is designed utilizing the proposed pass transistor logic-based PFD. The synthesizer boasts reduced phase noise, low power usage, high speed, and a broad frequency range of 0.1 to 4.75 GHz. The effectiveness of the suggested circuit was assessed across various supply and temperature conditions and all process corners.

A summary of phase frequency detectors and frequency synthesizers is outlined in Section 1. Section 2 demonstrates the working of the proposed PFD. A comprehensive description of the design optimization procedure utilizing Taguchi and ANOVA statistical approaches is given in Section 3. The analyses were carried out using Minitab software. The design and working of the frequency synthesizer, including its components like PFD, CP, LP, level shifter, DRO, and divider, are detailed in Section 4. Section 5 give details of the timing waveforms of the proposed PFD circuit. The characteristics of the differential ring oscillator, such as its oscillation frequency and oscillator gain w.r.t. control and supply voltage are also presented here. This section also showcases the dynamic behavior of the frequency synthesizer under different conditions and process variations, as well as its performance compared to previous research. The conclusion can be found in Section 6.

2. Proposed Phase Frequency Detector

The performance of a frequency synthesizer, encompassing factors such as speed, locking capability, and noise characteristics, is significantly impacted by the precise detection of the phase difference by the Phase Frequency Detector [28]. If the phase frequency detector (PFD) cannot discern minor phase differences, it results in dead zone at the output as illustrated in Fig. 2. As a consequence, the frequency synthesizer locks onto an incorrect phase. Consequently, it is imperative to adeptly tackle the dead zone issue [29].

Studies such as [30,31] indicate that the performance of the PFD is notably influenced by the blind zone. As the phase error reaches 360°, the PFD output is impacted by the presence of the blind zone. Although the optimal detection range of PFD is −360° to 360°, the occurrence of a blind zone causes it to shrink to $-360° + \delta$ to $360° - \delta$, where δ is calculated using Eq. (1). The effect of blind zone on the output is seen in Fig. 3.

$$\delta = 2\pi \frac{T_{rst}}{T_{ref}}. \tag{1}$$

The PFD uses T_{rst} to represent the pulse width of the reset signal and T_{ref} to represent the period of the reference input.

The blind zone can cause cycle sliding in the phase-locked loop (PLL) lock acquisition process, resulting in an extended lock time for the PLL. Designing an effective PFD block is a challenging undertaking because eliminating the dead zone and blind zone necessitates particular techniques that are restricted by the maximum operating frequency. Various methods have been suggested by researchers in the literature to mitigate these issues [30,31].

The PFD's ability to operate efficiently is significantly impacted by noise. Phase noise refers to the random fluctuations in a signal's phase, while jitter refers to the representation of phase noise in the time domain, resulting in unexpected rapid oscillations in the wave phase [32]. The power consumed by the Phase Frequency Detector (PFD) is a crucial contributor to a PLL's overall power usage. Thus, effectively managing the power consumed by the PFD can contribute to the overall management of the PLL's power consumption [33].

The traditional PFD design shown in Fig. 4(a) possesses an infinite capture range. Nevertheless, when input frequencies draw near to each other, the DFF's undergo a reset before any charge is transferred onto the filter capacitor by the CP. This phenomenon defines the dead zone. The existence of this dead zone results in jitter within the locked PLL. The presence of a large number of transistors contributes to an expanded layout area.

The proposed PFD shown in Fig. 4(b) has a unique structure that enables it to detect the rising edge of input signals. This is achieved through a two-stage design, which ensures sensitivity to rising edges by keeping the output of the first stage in its previous logic level when the two input signals have different levels. As a result, the PFD can detect input signal changes within the range of $[-\pi, \pi]$.

To better comprehend the performance of the proposed PFD, we can describe its working here: initially, both inputs of the PFD (C_{CLK} and C_{DOUT}) are set to a low level (zero logic). At this point, M1 and M2 are turned on, and the output of the first stage reaches a high level (One logic). At this point, the UP and DN signals are connected to the input signals through pass transistors M5 and M7, causing UP and DN to go to a low level. When C_{CLK} (C_{DOUT}) goes to a high logic level, the UP (DN) signal goes to a high level via M5 (M7), while the DN (UP) signal goes to a low level via M8 (M6). In the case where both inputs are high, UP and DN go to a low level via M6 and M8 without any delay for reset.

By eliminating the feedback path, this novel PFD can eliminate the dead zone and increase its operating speed, making it ideal for use in low-power and high-speed frequency synthesizers. Furthermore, the pass transistor logic can enhance the frequency of operation and simplify circuit implementation. The new structure of the PFD is constructed using 8 transistors, which results in reduced layout area and power usage of the PFD.

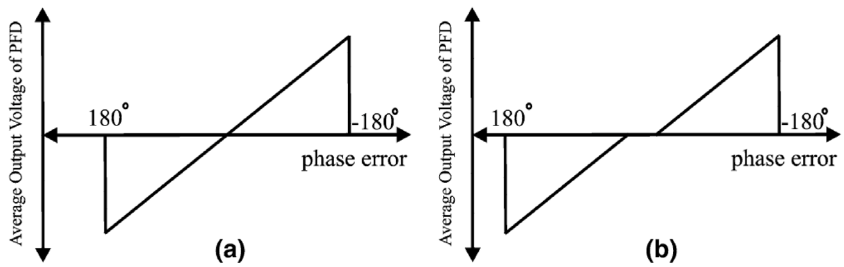


Fig. 2. Output of phase frequency detector (a) when dead zone is not present (b) when dead zone is present.

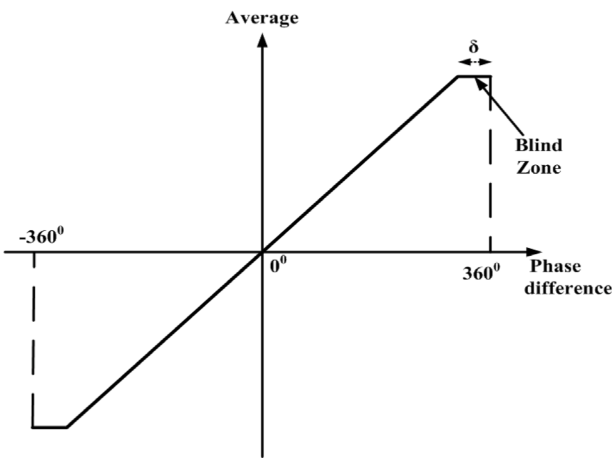


Fig. 3. The impact of the blind zone in PFD output.

3. Taguchi DoE and ANOVA statistical techniques for optimization of proposed PFD

Taguchi DoE (Design of Experiments) is a statistical approach for optimizing product or process design by identifying optimal parameter settings to improve product quality and reduce variability. It uses orthogonal arrays to efficiently evaluate the effects of multiple factors on a system's performance. The Taguchi method focuses on improving product or process robustness by identifying the most favorable combination of input parameters that result in the desired output [34,35]. The approach involves three stages:

• Design of orthogonal arrays: In this stage, an orthogonal array is designed based on the number of input parameters and their levels.

• Conducting experiments: Experiments are conducted by varying the input parameters at their respective levels, based on the orthogonal array design.

• Analysis of results: The experimental results are analyzed using ANOVA (Analysis of Variance) to determine the most significant input parameters and their optimal levels that affect the system's output.

The Taguchi DoE and ANOVA approach provides an efficient and systematic method to identify the most favorable parameter settings for improving product or process quality and reducing variability.

In this study, three independent variables were considered: the supply voltage (VDD), the width of PMOS transistors (W_p), and the width of NMOS transistors (W_n), with three levels for each control parameter. Table 1 presents these control parameters and their corresponding levels. Table 2 shows 9 pairs of experiments based on the L9 array. This approach permits the identification of the most favorable solution.

The Taguchi design of experiments (DoE) methodology suggests assessing the performance of the system by utilizing Signal to Noise Ratio (SNR) as a performance measurement metric. It is important to note that the SNR used in this context is distinct from the SNR that we use while designing the analog circuits. In this section, signal to noise ratio has been calculated for each experiment and output parameter and the corresponding SNR values will be determined based on the required output performance parameter. The SNR for the smaller the better condition and larger the better condition is given in Eqs. (2) and (3) respectively.

Table 1
Control parameters and their levels.

Sr.	Control parameter	Level 1	Level 2	Level 3
1.	VDD (mV)	1.2	1.5	1.8
2.	W_p (μm)	0.42	0.5	0.58
3.	W_n (μm)	0.42	0.5	0.58

$$SNR_{smaller} = -10 \times log_{10}\left(\frac{1}{n}\sum_{i=1}^{n} {Y_i}^2\right) \tag{2}$$

$$SNR_{larger} = -10 \times log_{10}\left(\frac{1}{n}\sum_{i=1}^{n} \frac{1}{{Y_i}^2}\right) \tag{3}$$

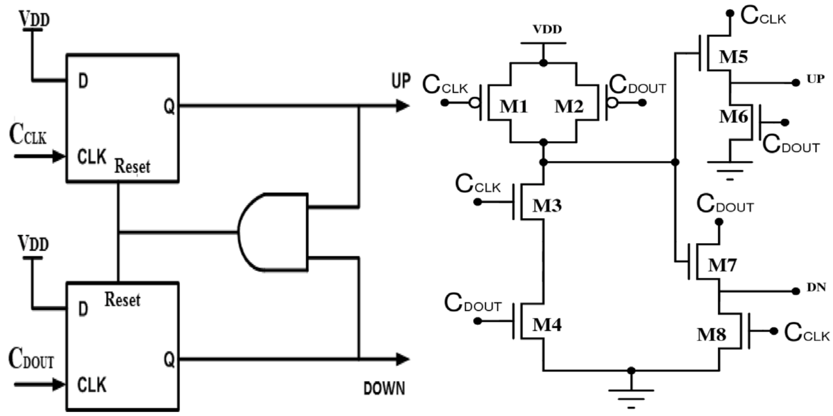
where Y_i represents the output value and n is the total number of experiments. In Figs. 5(a)–5(d), the plots of SNR for different outputs are presented.

The greatest impact on response corresponds to the largest value of SNR, and vice versa. This study selects four output parameters to develop an optimum solution: phase noise, power dissipation, maximum operating frequency, and delay. The low power, low delay, low phase noise, and high maximum operating frequency are taken into consideration to calculate the corresponding SNR. The phase noise has a negative value so to simplify the calculations, the negative sign has been removed and SNR calculation has been done considering the high phase noise. Minitab software was used to conduct this statistical study. Table 2 displays the output values of various parameters and their corresponding SNR values. Experiment number 9 resulted in the highest phase noise, power consumption, and maximum operating frequency with a high SNR value while achieving the lowest delay with the lowest SNR value.

Table 3 uses "Delta" to represent the impact of the control parameter on the performance of the corresponding performance parameter. A larger Delta value suggests that the control factor has a stronger impact on the associated performance parameter. Based on the information presented in Figs. 5(a)–5(d) and Table 3, it can be deduced that VDD has a greater influence on all PFD parameters compared to W_p and W_n, as indicated by its higher Delta value.

Table 3 is used to calculate the average SNR of all output parameters (phase noise, power consumption, maximum operating frequency, and delay) at 3 different levels for each control variable (VDD, W_p, and W_n). The level corresponding to the highest average SNR for each control variable is selected. Based on these calculations, the levels of VDD, W_p, and W_n are determined to be 3, 1, and 3, respectively, corresponding to values of 1.8 V, 0.42 μm, and 0.58 μm.

To determine which independent variable or control factor has the most significant impact on a specific output parameter of the PFD, ANOVA testing is performed. The *P*-value obtained from this test can be used to identify the control factor that has the greatest influence

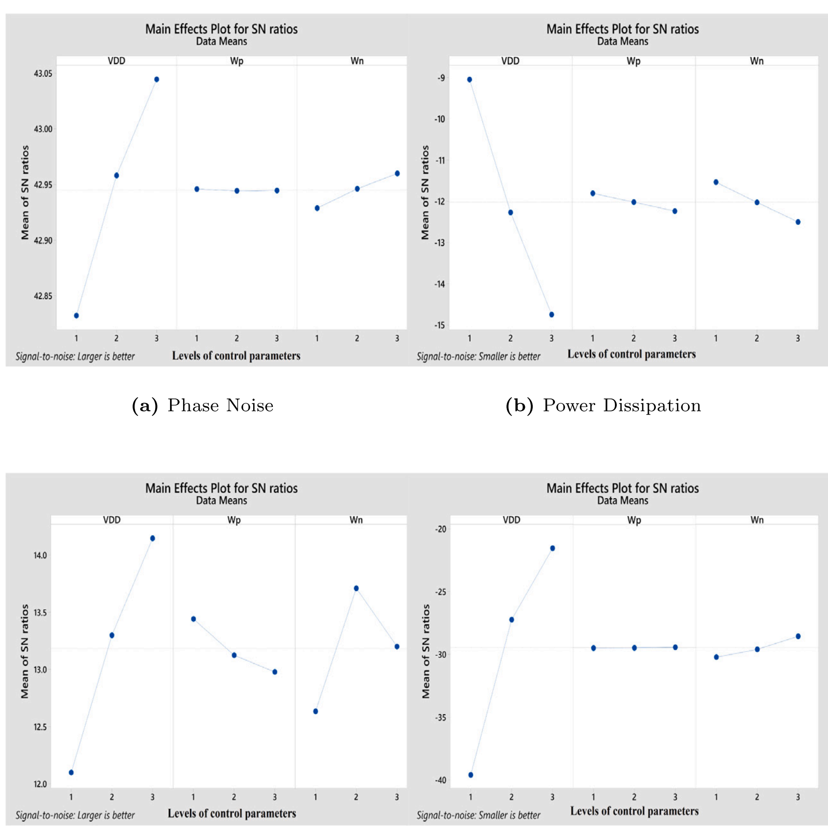


(a) Traditional PFD Circuit (b) Proposed PFD Circuit

Fig. 4. Traditional and proposed phase frequency detectors.

Table 2
Outputs and corresponding signal-to-noise ratios of proposed PFD.

S.No.	VDD	W_p	W_n	PNoise (dBc/Hz)	SNR (PNoise)	Power (μW)	SNR (Power)	F_{max} (GHz)	SNR (F_{max})	Delay (ps)	SNR (Delay)
1.	1	1	1	−138.322	−42.818	2.610	−8.333	4.02	12.085	101.9	−40.164
2.	1	2	2	−138.564	−42.833	2.835	−9.051	4.24	12.547	95.89	−39.636
3.	1	3	3	−138.771	−42.846	3.069	−9.739	3.83	11.664	89.2	−39.007
4.	2	1	2	−140.592	−42.959	4.008	−12.059	4.92	13.839	24.38	−27.741
5.	2	2	3	−140.819	−42.973	4.335	−12.739	4.75	13.534	20.22	−26.116
6.	2	3	1	−140.311	−42.942	3.983	−12.004	4.23	12.527	24.68	−27.847
7.	3	1	3	−142.243	−42.061	5.635	−15.018	5.25	14.403	10.65	−20.547
8.	3	2	1	−141.687	−43.027	5.166	−14.263	4.62	13.293	13.58	−22.658
9.	3	3	2	−142.007	−43.046	5.599	−14.962	5.46	14.744	11.79	−21.430



(a) Phase Noise (b) Power Dissipation

(c) Max. Operating Frequency (d) Delay

Fig. 5. SNR ratio graphs of the proposed PFD circuit.

Table 3
SNR response table of various parameters.

(a) Phase noise

Level	VDD	W_p	W_n
1	42.83	42.95	42.93
2	42.96	42.94	42.95
3	42.71	42.94	42.96
Delta	0.25	0.01	0.03
Rank	1	3	2

(b) Max. operating frequency

Level	VDD	W_p	W_n
1	12.10	13.44	12.63
2	13.30	13.12	13.71
3	14.15	12.98	13.20
Delta	2.05	0.46	1.08
Rank	1	3	2

(c) Power dissipation

Level	VDD	W_p	W_n
1	−9.041	−11.803	−11.533
2	−12.268	−12.018	−12.024
3	−14.748	−12.235	−12.499
Delta	5.706	0.432	0.966
Rank	1	3	2

(d) Delay

Level	VDD	W_p	W_n
1	−39.6	−29.48	−30.22
2	−27.23	−29.47	−29.60
3	−21.55	−29.43	−28.56
Delta	18.01	0.06	1.67
Rank	1	3	2

Table 4
ANOVA test for targeted output parameters.

(a) Phase noise

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	17.9992	5.9997	154.69	0.000
VDD	1	43.3091	43.3091	67.85	0.000
W_p	1	0.0008	0.0008	0.02	0.893
W_n	1	0.3816	0.3816	9.84	0.026
Error	5	0.1939	0.0388		
Total	8	18.1931			

(b) Power dissipation

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	10.6643	3.5548	769.38	0.000
VDD	1	10.3648	10.3648	2243.32	0.000
W_p	1	0.0264	0.0264	5.71	0.062
W_n	1	0.2731	0.2731	59.10	0.001
Error	5	0.0231	0.0046		
Total	8	10.6874			

(c) Max. operating frequency

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	1.97802	0.65934	6.53	0.035
VDD	1	1.74960	1.74960	17.34	0.009
W_p	1	0.07482	0.07482	0.74	0.429
W_n	1	0.15360	0.15360	1.52	0.272
Error	5	0.50447	0.10089		
Total	8	2.48249			

(d) Delay

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	3	10 586.1	3528.7	9.29	0.017
VDD	1	10 497.7	10 497.7	27.64	0.003
W_p	1	21.1	21.1	0.06	0.823
W_n	1	67.3	67.3	0.18	0.691
Error	5	1899.2 379.8			
Total	8	12 485.3			

Table 5
Optimized solution and predictable output values from ANOVA.

VDD (V)	W_p (μm)	W_n (μm)	Phase noise (dBc/Hz)	Power Diss. (μW)	Fmax (GHz)	Delay (ps)
1.8	0.42	0.58	−139.69	5.47	5.52	9.97

on an output parameter. The control factor with the lowest P-value is considered to have the strongest impact on the circuit's output parameter.

Table 4 displays the general linear models created using ANOVA for the output characteristics. The analysis indicates that VDD has the lowest P-value among all control factors for all parameters listed in the table. This finding implies that VDD is a crucial factor in optimizing the performance of all parameters. Eqs. (4)–(7) are derived from the ANOVA analysis and represent the relationship between the input variables and the performance parameters.

$$Phasenoise = 136.460 + 1.7135VDD - 0.0114W_p + 0.2522W_n \tag{4}$$

$$PowerDissipation = 2.95 + 1.3143VDD + 0.0663W_p + 0.2133W_n \tag{5}$$

$$F_{max} = 4.5 + 0.540VDD - 0.112W_p + 0.160W_n \tag{6}$$

$$Delay = 88 - 41.83\ VDD - 1.88W_p - 3.35W_n \tag{7}$$

To obtain the best possible performance in terms of maximum operating frequency, phase noise, delay, and power consumption, the proposed PFD has been optimized using ANOVA. Table 5 shows the predicted results obtained through this method.

The values of VDD, W_p, and W_n obtained from ANOVA are used to design and simulate the proposed PFD to validate the reliability of the Taguchi DoE and ANOVA. The results of the simulation show that the PFD has a phase noise of −142.24 dBc/Hz, a power consumption of 5.64 μW, an operating frequency of 5.25 GHz, and a delay of 10.65 ps.

Table 6
Comparison of proposed PFD results with predicted solution from Taguchi DoE and ANOVA.

Parameters	Predicted values	Simulated values	% deviation
Phase noise (dBc/Hz)	−139.69	−142.24	1.83
Power consumption (μW)	5.47	5.64	3.11
Max. Op. frequency (GHz)	5.52	5.25	4.9
Delay (ps)	9.97	10.65	6.82

Table 6 displays a comparison of the performance of the proposed PFD with the predicted solution obtained through the Taguchi DoE and ANOVA statistical approach. The results indicate that the simulated values are just as accurate as the predicted values.

4. Application of the proposed PFD in frequency synthesizer

The proposed optimized PFD circuit is used to design a novel frequency synthesizer, which is high-speed, low noise, and compact. The proposed PFD provides enhanced phase and frequency error tracking, improved noise performance for high-frequency applications, and faster lock time, thus making it a suitable option for fast-locking systems. The proposed synthesizer is depicted in Fig. 6. The literature describes various PLL-based synthesizers to generate a wide range of frequencies. The feedback division ratio affects the generated frequency.

The first component of the suggested synthesizer is the PFD, depicted in Fig. 4. The phase/frequency of the clock input is compared to the phase/frequency of the feedback output from the divider by PFD.

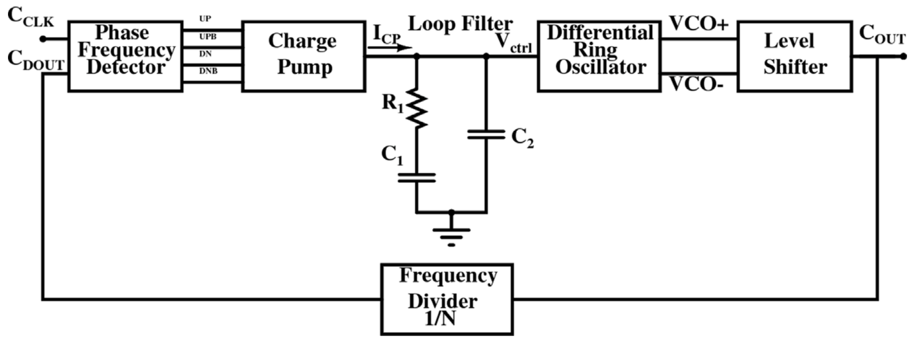


Fig. 6. Proposed frequency synthesizer showing all the sub-blocks.

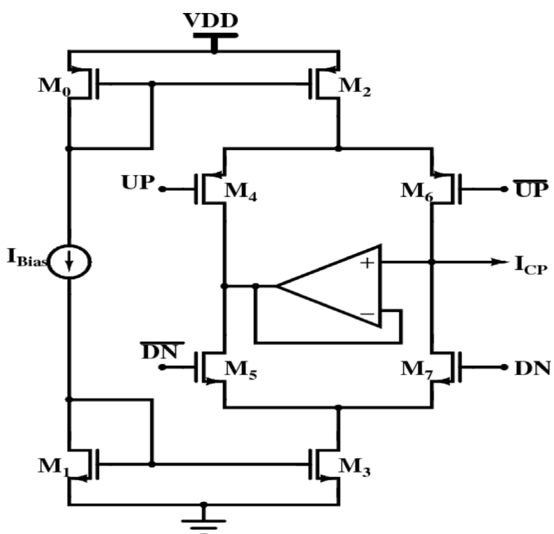


Fig. 7. Circuit diagram of charge pump [36].

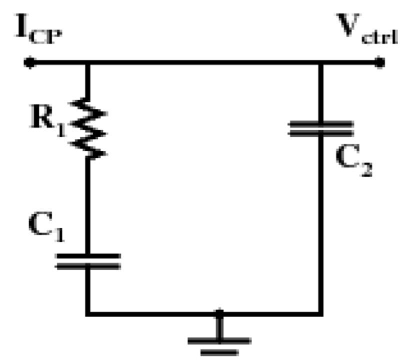


Fig. 8. Second-order loop filter.

Depending on whether the frequency or phase of the feedback signal (C_{Dout}) is higher or lower than the frequency or phase of the clock signal (C_{CLK}), the PFD produces four outputs that are processed by the Charge Pump (CP) to provide a control signal generated through the loop filter. This signal is then sent to the differential ring oscillator.

The charge pump plays a crucial role in the phase-locked loop (PLL) circuit within a frequency synthesizer, responsible for regulating the frequency of the output signal. The quality of the charge pump design can greatly affect the overall stability and noise performance of the PLL circuit and, as a result, the frequency synthesizer as a whole. The main sources of noise in CP are mismatched up and down current and uneven transistor switching resistance. These inconsistencies result in a static phase offset error, causing ripples in control voltage and negatively impacting performance in terms of phase noise. In the synthesizer being studied, a bootstrapped charge pump is employed for its low swing operation and differential current steering. To ensure equal up and down current, a voltage follower operational amplifier is used. The charge pump schematic is depicted in Fig. 7.

The output current I_{CP} from the charge pump is converted to a control voltage named V_{ctrl} with the help of a loop filter (LF). The LF is a crucial component in the proposed circuit, with its characteristics determining the design parameters of the synthesizer. The circuit for the loop filter is depicted in Fig. 8.

Eq. (8) represents the transfer function of the loop filter.

$$F(s) = \left(R_1 + \frac{1}{C_1 s}\right) \parallel \left(\frac{1}{C_2 s}\right) \tag{8}$$

To simplify the calculations, the capacitor C_2 can be disregarded, and the transfer function is presented by Eq. (9).

$$F(s) = \left(R_1 + \frac{1}{C_1 s}\right) \tag{9}$$

Eq. (10) represents the transfer function of the synthesizer in Fig. 6.

$$\frac{\phi_D}{\phi_R} = \frac{\frac{I_{CP}K_{DRO}}{2\pi N}\left(R_1 + \frac{1}{sC_1}\right)}{s + \frac{IK_{DRO}}{2\pi N}\left(R_1 + \frac{1}{sC_1}\right)} = \frac{\frac{IK_{DRO}}{2\pi NC_1}\left(sR_1C_1 + 1\right)}{s^2 + \frac{IK_{DRO}}{2\pi N}R_1 s + \frac{IK_{VCO}}{2\pi NC_1}} \tag{10}$$

The phase of the divider and reference signal are denoted as ϕ_D and ϕ_R, respectively. The DRO gain is represented by K_{DRO}, and the current from the charge pump is I_{CP}. The divider ratio is symbolized as N, and the loop filter components are R_1, C_1, and C_2.

The natural frequency and damping constant of the loop are respectively represented by Eqs. (11) and (12).

$$\omega_n = \sqrt{\frac{I_{CP}K_{DRO}}{2\pi NC_1}} \tag{11}$$

$$\zeta = \frac{R_1}{2}\sqrt{\frac{I_{CP}K_{DRO}C_1}{2\pi N}} \tag{12}$$

The passive elements in the loop filter are commonly determined based on the damping constant and natural frequency. Eqs. (13), (14), and (15) can be used to calculate the values of C_1, R_1, and C_2, respectively.

$$C_1 = \frac{I_{CP}K_{DRO}}{2\pi N\omega_n^2} \tag{13}$$

$$R_1 = 2\zeta\sqrt{\frac{2\pi N}{I_{CP}K_{DRO}C_1}} \tag{14}$$

$$C_2 = \frac{C_1}{10} \tag{15}$$

For simplicity, C_2 is set to one-tenth of C_1.

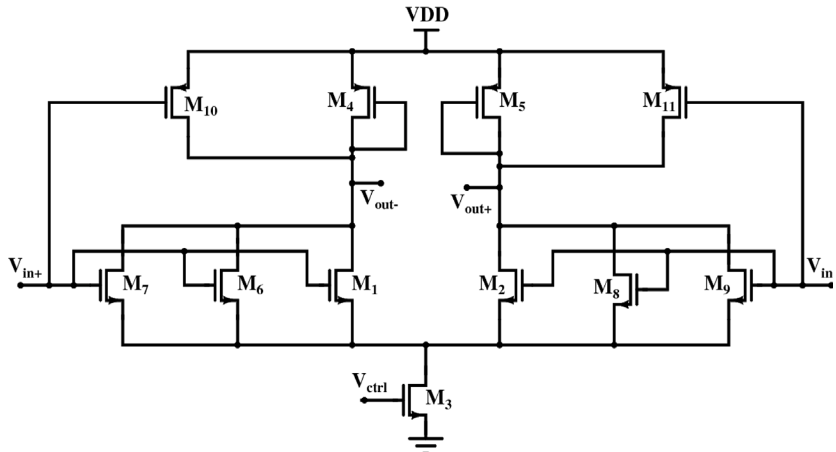


Fig. 9. Delay cell for differential ring oscillator.

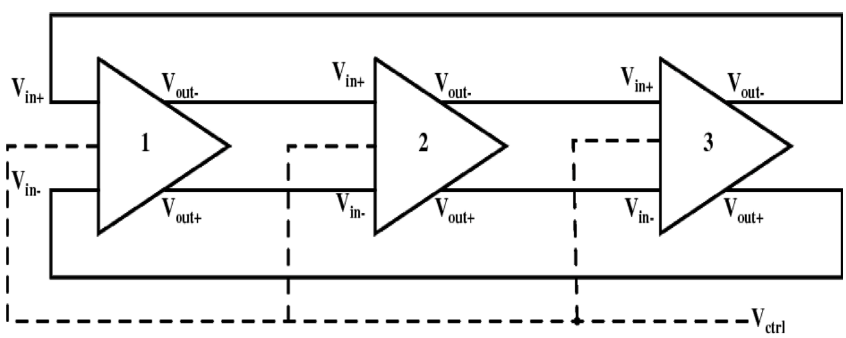


Fig. 10. 3-stage differential ring oscillator.

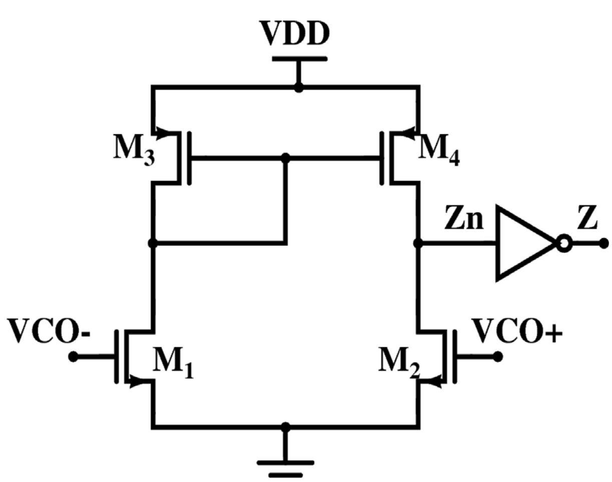


Fig. 11. Circuit diagram of level shifter.

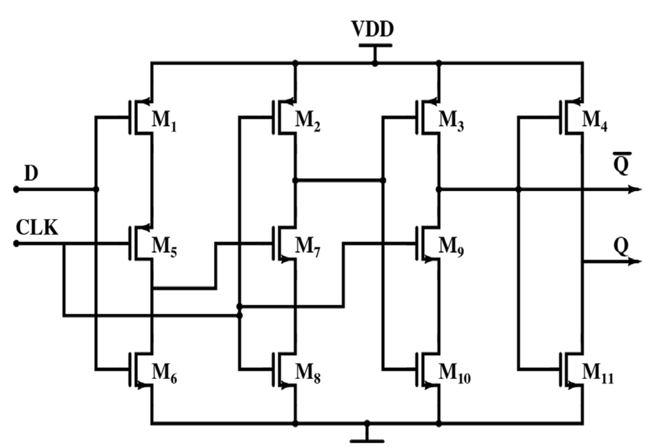


Fig. 12. D flip flop schematic used in the divider circuit.

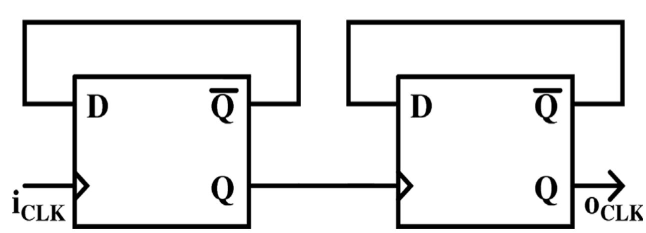


Fig. 13. Divide-by-4 divider circuit.

The output of the loop filter in the form of the control voltage (V_{ctrl}) is used to control the frequency of oscillation of the differential ring oscillator (DRO) of Fig. 10. The delay circuit, depicted in Fig. 9, plays a crucial role in the DRO. An odd or even number of cascaded delay cell stages can be used to construct a DRO circuit. In this study, three cascaded delay cells were used to create a 3-stage DRO [37]. PMOS transistors include M_4, M_5, M_{10}, and M_{11}, while $M_1, M_2, M_6, M_7, M_8, M_9$, and M_3 are NMOS transistors. The differential inputs are V_{in+} and V_{in-}, while the differential outputs are V_{out+} and V_{out-}. The oscillation frequency of the DRO can be controlled by adjusting the voltage V_{ctrl} applied to the MOSFET M_3. The level shifter

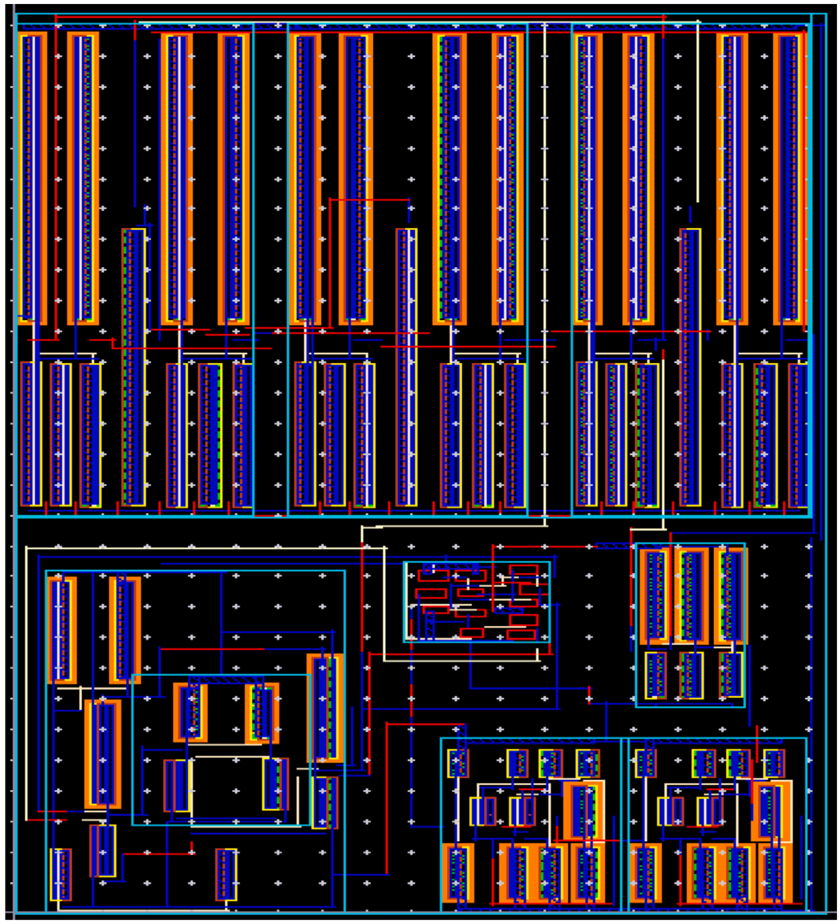


Fig. 14. Layout of the proposed frequency synthesizer.

circuit depicted in Fig. 11 is used to amplify and convert the differential output from the DRO into a single-ended signal.

Next, the output of the DRO after being passed through the level shifter is sent to the input of a frequency divider, which transforms the signal to a lower frequency and then the signal from the divider is sent back to PFD so that it can be compared with the reference signal. In the synthesizer, a divide-by-4 divider is used for this purpose. The divider is composed of two D flip-flop circuits depicted in Fig. 12. Fig. 13 depicts a block diagram of the divider.

A visual representation of the layout of the proposed high-speed frequency synthesizer can be seen in Fig. 14. The area occupied by the chip measures 0.013 square millimeters.

5. Results and discussion

In this work, a new pass transistor-based PFD shown in Fig. 4 is presented. The PFD is designed using CADENCE Virtuoso and operates on a 1.8 V supply. The transient behavior of the proposed PFD is displayed in the timing diagram of Fig. 15. The diagram shows that the DN signal goes high and the UP signal goes low when the phase of the feedback signal from the divider (C_{DOUT}) leads the input clock (C_{CLK}). When the input clock leads the divider output, the UP signal becomes high.

Fig. 16(a) shows the phase noise of the proposed PFD is shown to be −142.24 dBc/Hz at a 1 MHz offset. Fig. 16(b) demonstrates that the linear characteristic of the PFD which is due to the elimination of the dead zone. The maximum operating frequency of the PFD is determined to be 5.25 GHz, using the technique suggested in [38]. The proposed PFD design comprises of eight transistors with no reset path, and has a power usage of 5.64 μW at 5.25 GHz.

The DRO utilized in the synthesizer has a tuning range of 0.1 to 4.75 GHz. A control voltage generated with the help of CP and LF is passed through DRO to generate the desired frequency range. When the control voltage is varied from 0 V to 1.8 V, the above-mentioned range of frequencies is obtained. The gain and frequency characteristics of the DRO are shown in Fig. 17 with respect to control voltage and supply voltage.

In the proposed high-speed, low-power CP-PLL frequency synthesizer, a divider ratio of N = 4 is chosen to produce a frequency of 4 GHz, with a reference input frequency of 1 GHz. The values of resistance R = 240 Ω and capacitances C_1 = 2.65 nF, C_2 = 0.265 nF for the loop filter are calculated using Eqs. (7) to (9). The transient response of the synthesizer is shown in Fig. 18(a), where it can be seen that the lock occurs at 2.95 μs and the control voltage V_{ctrl} stabilizes at 1.02 V, leading to an output frequency of 4 GHz from the DRO. Fig. 18(b) shows the zoomed output waveforms of the frequency synthesizer.

The frequency range in which the synthesizer locks can be seen from Fig. 19 which is a plot of frequency versus time. The locking range is from 3.98 GHz to 4.02 GHz.

The outcomes of the analysis on Process, Voltage, and Temperature (PVT) variations for the suggested frequency synthesizer are presented in Table 7, which includes both pre-layout and post-layout simulation data. The findings suggest that alterations in the supply voltage only result in a slower locking process, while the frequency output remains unaffected. The proposed frequency synthesizer is capable of generating frequencies between 0.1 and 4.75 GHz, with a power usage of 4.25 mW while producing a frequency of 4 GHz. Performance of the synthesizer is analyzed and recorded in Table 8.

In this work, a novel figure of merit (FoM) has been introduced that takes into account various factors such as phase noise ($PN(\Delta f)$),

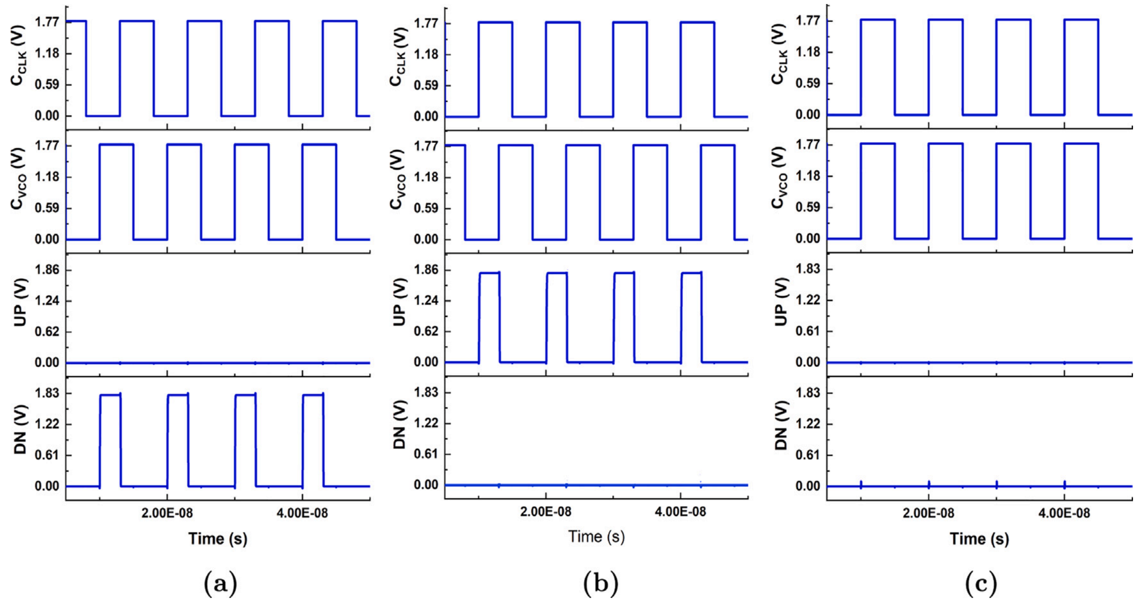


Fig. 15. Transient behavior of phase frequency detector circuit (a) C_{CLK} leads C_{DOUT} (b) C_{CLK} lags C_{DOUT} (c) C_{CLK} in-phase with C_{DOUT}.

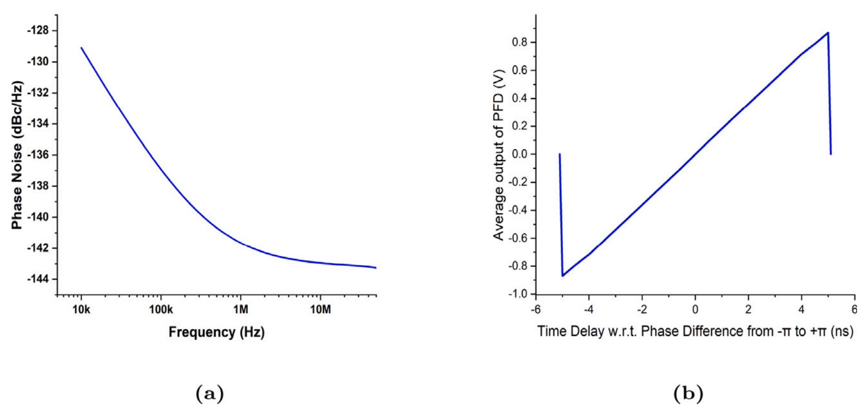


Fig. 16. (a) Phase noise characteristics of the proposed PFD (b) ϕ-V Characteristics of the proposed PFD.

Table 7
Performance parameters of a frequency synthesizer with respect to Process, Voltage, and Temperature (PVT) variations.

Temp = 27 °C	Supply Voltage Variation					
	Pre layout			Post layout		
Supply voltage (V)	Lock time (μs)	Power dissipation (mW)	THD %	Lock time (μs)	Power dissipation (mW)	THD %
1.71	2.54	4.13	7.76	2.63	4.78	8.03
1.8	2.95	4.25	13.23	3.12	4.89	13.79
1.89	6.43	5.06	16.16	7.35	5.34	17.34
VDD = 1.8 V	Process variation					
	Pre layout			Post layout		
FF	2.32	4.10	12.44	2.65	5.01	12.77
FS	5.98	4.15	15.24	6.21	5.09	15.86
NN	2.95	4.25	13.23	3.12	5.33	13.79
SF	6.03	4.27	16.14	6.67	5.45	16.96
SS	9.21	4.30	9.23	10.04	5.62	10.21
VDD = 1.8 V	Temperature variation					
	Pre layout			Post layout		
−30 °C	3.13	4.78	10.23	4.21	5.16	10.54
0 °C	2.64	4.85	14.35	3.13	5.24	15.23
85 °C	2.36	5.02	7.01	3.05	5.53	7.65

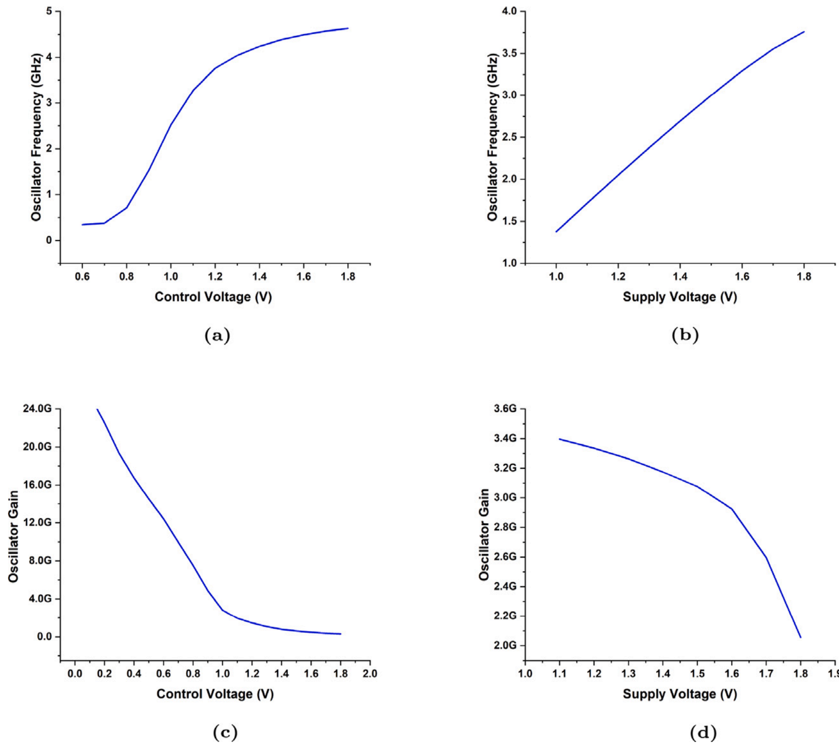


Fig. 17. Transfer characteristics of DRO (a) Frequency vs. Control voltage (b) Frequency vs. Supply voltage (c) Gain vs. Control voltage (d) Gain vs. Supply voltage.

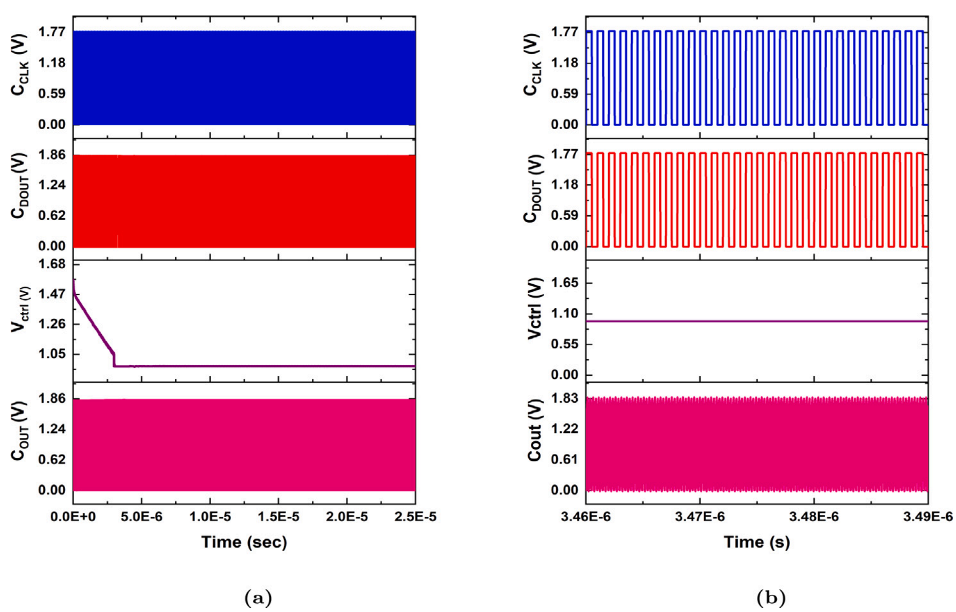


Fig. 18. Transient response of frequency synthesizer (a) original response (b) zoomed response to show locking phenomenon.

Table 8
Performance Summary of Proposed Frequency Synthesizer.

Parameter	Value
Supply voltage	1.8 V
Reference frequency	1 GHz
Loop bandwidth	1 MHz
Charge pump current	200 μA
Setting time	2.95 μs
Jitter	212 ps
Power dissipation	4.25 mW
DRO gain (K_{DRO})	3.289 GHz/V
Output frequency range	100 MHz–4.75 GHz
Locking frequency range	3.98 GHz–4.02 GHz

Table 9
Comparison of the proposed frequency synthesizer with the work done in literature.

Ref.	Tech. (nm)	PS (V)	OF (GHz)	FR (GHz)	LT (μs)	PN (dBc/Hz)	PD (mW)	Area (mm^2)	FoM (dB)
[2]	180	1.8	1.6	0.025–1.6	2.5	−137	2.63	0.013	−155.71
[19]	180	1.8	2.56	0.2–4.27	2.95	−113.64	4.9	0.154	−130.52
[20]	45	1	2.4	2–3	0.2	−113.8	4	0.015	−140.61
[24]	65	1.1	2.2	2.05–2.2	4	−122	8.8	0.24	−103.11
[25]	180	1.8	2.4	2.11–2.42	35	−107.4	21.3	0.744	−77.39
[26]	130	1.2	3.8	1.9–3.8	22	−117.5	12.8	1.8	−105.07
[27]	180	1.8	2.1	1.6–2.1	15	−105	20.5	0.42	−76.78
[39]	130	1.2	4.5	0.9–5.4	3	−119	24	0.8	−127.52
[40]	180	1	2.44	4.56–5.32	60	−121.9	9.1	1.33	−98.59
[41]	180	1.8	2.431	2.368–2.496	20	−113	29.6	2.1	−71.91
[42]	180	1.8	1.72	1.72–1.74	18	−119.3	10.6	2.4	−63.42
This work	180	**1.8**	**4**	**0.1–4.75**	**2.95**	**−142.23**	**4.25**	**0.013**	**−175.24**

Tech: Technology, PS: Power Supply, OF: Output Frequency, FR: Frequency Range, LT: Locking Time, PD: Power Dissipation, FoM: Figure of Merit.

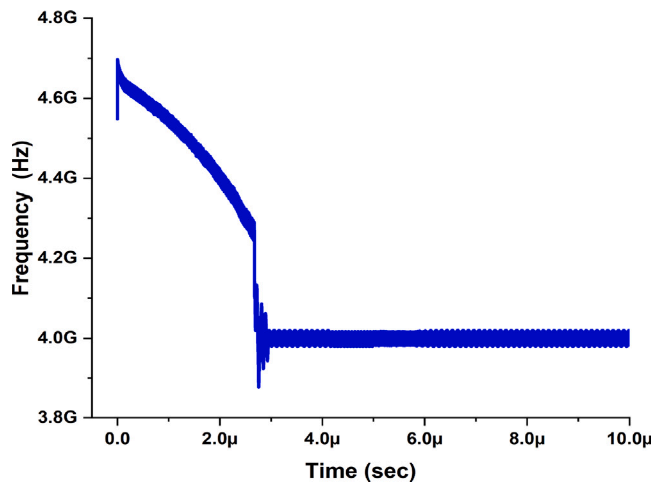


Fig. 19. Frequency vs. Time response of frequency synthesizer.

oscillation frequency (f_{osc}), tuning range (with f_{max} and f_{min} being the maximum and minimum frequencies), power dissipation (PD), chip area occupied (A), and settling time (t_{set}). This figure of merit is calculated using Eq. (16). As shown in Table 9, the FoM improves as the output frequency and tuning range increase.

$$FOM_{new} = PN(\Delta f) - 10log\left[\frac{\left(\frac{f_{osc}}{\Delta f}\right)^2\left(\frac{f_{max}-f_{min}}{1\ \mathrm{GHz}}\right)^2\left(\frac{1\ \mathrm{mW}}{PD}\right)}{A.t_{set}}\right] \quad (16)$$

The phase noise at a specific frequency of interest is represented by $PN(\Delta f)$, the oscillation frequency is denoted as f_{osc}, and the maximum and minimum frequencies in the tuning range are represented by f_{max} and f_{min} respectively, power dissipation is represented by PD, the area occupied on the chip is represented by A, and the settling time is represented by t_{set}.

Table 9 displays a comparison of the proposed frequency synthesizer with the work done in earlier research. This synthesizer has the broadest tuning range, lowest power consumption, smaller chip area, and quickest settling time, as indicated by the comparison table.

6. Conclusion

In this work, a pass transistor-based phase frequency detector is used to design a new high-speed, low-power, CP-PLL-based synthesizer. The use of a pass transistor-based PFD enhances the performance and accuracy of the synthesizer through improved phase and frequency error tracking, higher noise performance for high-frequency applications, and faster lock time. This work is accomplished in CADENCE Virtuoso using CMOS 180 nm GPDK and 1.8 V supply. This frequency synthesizer can produce frequencies between 0.1 to 4.75 GHz. Its performance, including lock time, THD, and power consumption, was evaluated under varying temperature and supply voltage conditions and different process corners. The synthesizer uses just 4.25 mW and has a small layout of 0.013 mm^2. Additionally, the synthesizer boasts a settling time of 2.95 μs. A novel figure of merit, FoM, was established and compared with other synthesizers in the literature. The proposed synthesizer achieved a figure of merit of −175.24 dB, outperforming other circuits and making it suitable for use in high-speed devices such as radio receivers, televisions, mobile phones, satellite receivers, and GPS systems.

Declaration of competing interest

The authors hereby declare that there exists no conflict of interest in this work, or in publishing this work in the Integration.

Data availability

Data will be made available on request.

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